



Averages

AN AVERAGE IS A "MIDDLE" VALUE OF A SET OF DATA. IT IS A TYPICAL VALUE THAT REPRESENTS THE ENTIRE SET OF DATA.

Different types of averages

There are several different types of average. The main ones are called the mean, the median, and the mode. Each one gives slightly different information about the data. In everyday life, the term "average" usually refers to the mean.

The mode

The mode is the value that appears most frequently in a set of data. It is easier to find the mode if you put the data list into an ascending order of values (from lowest to highest). If different values appear the same number of times, there may be more than one mode.

150, 160, 170, 180, 180

this color represents mode because it appears most often



working out averages often requires listing a set of data arranged in ascending order

◀ The mode color

The set of data in this example is a series of colored figures. The pink people appear the most often, so pink is the mode value.

150, 160, 170, 180, 180

180 occurs twice in this list, more often than any other value, so it is the mode, or most frequent, value

▷ Average heights

The heights of this group of people can be arranged as a list of data. From this list, the different types of average can be found—mean, median, and mode.

mode the value that appears most often in the list

median the middle value of the list

mean the sum of the values divided by the number of values in the list (168 cm)

this person is 170 cm tall; 170 cm is the median (middle value in the list)

this person is one of two people who are 180 cm tall; 180 cm is therefore the mode value

shortest person is 150 cm tall

this person is 160 cm tall, shorter than average

